

Amartya Dutta

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RESEARCH FOCUS

My research focuses on drug repurposing using graph-based and multimodal language model (MLM)-driven approaches.

EDUCATION

Virginia Tech, Blacksburg, USA

PhD, Computer Science; GPA: 3.92/4

May 2025 – Present

Passed Qualifiers

MS, Computer Science; GPA: 3.9/4

Aug 2022 – Feb 2025

Thesis: [Zero-Shot Scene Graph Relationship Prediction using VLMs](#)

Indian Institute of Information Technology (IIIT) Guwahati, India

BTech, Computer Science and Engineering; GPA: 3.44/4

Jul 2017 – May 2021

WORK EXPERIENCE

AI Research Intern, Genentech

May 2026 – Present

- Researching novel approaches to building a Multimodal Biomedical Foundation Model.

Graduate Research Assistant, COMPASS Centre

May 2025 – May 2026

- Developing graph-based approaches for antiviral drug repurposing.
- Contributed to a virology-focused scientific information extraction project by benchmarking a curated virus-mutation dataset against state-of-the-art LLM-based RAG frameworks to assess extraction performance on domain-specific biomedical literature.

Graduate Research Assistant, Virginia Tech

Aug 2022 – Dec 2024

- Developed a **novel zero-shot approach** for **Scene Graph Relationship Prediction** using **VLMs**, reframing Predicate Classification as an **MCQA task** and surpassing trained baselines by at least 7% for **balanced relationship prediction**. Designed an **open-ended relationship generation framework** to eliminate answer-choice biases and improve contextual understanding.
- Performed comparative analysis of **Weakly Supervised Semantic Segmentation** approaches, highlighting the superiority of saliency maps over CAMs and introducing stochastic aggregation to enhance saliency effectiveness.

Augmented Reality Developer Intern, Amply

Dec 2019 – Mar 2020

- Developed interactive Augmented Reality (AR) portals using AR-Core in Unity3D, enabling secure interactions with virtual objects to create immersive AR tours for client companies.

Virtual Reality Developer Intern, IIT Guwahati

May 2019 – Jul 2019

- Designed and developed an interactive Virtual Reality (VR) tour using Unity3D, focusing on smooth navigation within the virtual environment to enhance user experience (UX) in VR.

PUBLICATIONS

- Blessy Antony, **Amartya Dutta**, Anuj Karpatne, T. Murali et al. "VILLA: Versatile Information retrieval from scientific Literature using large LLanguage models". [KDD'26: AI4Science](#). [Paper](#)
- Amartya Dutta**, M. Maruf, Ismini Lourentzou, Arka Daw, Anuj Karpatne et al. "Open World Scene Graph Generation using Vision Language Models". [CVPR'25 Workshop](#), [ICML'25 Workshop](#). [Paper](#)
- Abhilash Neog, Medha Sawhney, Kazi Sajeed Mehrab, **Amartya Dutta**, Anuj Karpatne et al. "Toward Scientific Foundation Models for Aquatic Ecosystems". [ICML'25 Workshop](#)
- Medha Sawhney, Abhilash Neog, Mridul Khurana, **Amartya Dutta**, Arka Daw, Anuj Karpatne. "Physics-guided Diffusion Neural Operators for Solving Forward and Inverse PDEs". [CVPR'25 Workshop Poster](#)
- Sepideh Fatemi, Abhilash Neog, Emma Marchisin, **Amartya Dutta**, Anuj Karpatne et al. "Scientific Equation Discovery using Modular Symbolic Regression via Vision-Language Guidance". [CVPR'25 Workshop Poster](#)
- M. Maruf, Arka Daw, **Amartya Dutta**, Jie Bu, Anuj Karpatne. "Beyond Discriminative Regions: Saliency Maps as Alternatives to CAMs." [Paper](#)
- Amartya Dutta**, Rajat Kanti Bhattacharjee, Ferdous Ahmed Barbhuiya. "Efficient Detection of Lesions During Endoscopy." [ICPR'21 Workshop](#). [Paper](#)
- Amartya Dutta**, Kamaljiyoti Nath. "Learning via LSTM for Railway Bridge strains." [ICDSMLA'20](#). [Paper](#)
- Amartya Dutta**, Ferdous Ahmed Barbhuiya. "Predicting Popularity of Images Over 30 Days." [Paper](#)

PROJECTS

- **SealQA:** Contributed towards the creation a dataset of complex questions that challenge state-of-the-art LLMs. This project evaluates how these models handle queries requiring up-to-date knowledge and complex reasoning by leveraging search engines for real-time information. [Paper](#)
- **Evaluating Model Reasoning and Hallucinations in Medical LLMs:** This project investigates factual error propagation in open-source medical LLMs (e.g., BioMistral, Asclepius) and documents their datasets for transparency. By highlighting performance variations, it aims to guide the development of safer, more reliable language models for healthcare. [GitHub](#)
- **Visualizing the Spotify Soundscape:** This project visualizes the Spotify Top 50 Tracks of 2023 through an interactive, HTML-based dashboard. Using D3.js and Plotly.js, it enables dynamic, data-driven exploration of each track's popularity and attributes. [GitHub](#)
- **Predicting Popularity of Flickr Images (ICIP 2021):** This project predicts how popular a Flickr image will be over 30 days, even before it's uploaded. By analyzing user and image social features alongside image visuals, it models engagement based on two factors: scale and shape. Using these factors, the method forecasts the daily engagement sequence. [GitHub](#)

TEACHING EXPERIENCE

- **CS 2064:** Intermediate Programming in Python – Graduate Teaching Assistant, Virginia Tech
- **CS 5644:** Machine Learning with Big Data – Graduate Teaching Assistant, Virginia Tech
- **CS 5805:** Machine Learning – Graduate Teaching Assistant, Virginia Tech

ACADEMIC SERVICE & AWARDS

- **Reviewer:** International Journal of Computer Vision; KDD'26: AI4Sciences Track; CVPR'25: CVinW Workshop; AAAI'24: Imageomics-Workshop
- **Program Committee:** CVPR 2025 Workshop, CVinW
- **3rd Position:** IEEE ICIP'20 Image Popularity Prediction Challenge [View Results](#)

SKILLS SUMMARY

- **Languages:** Python, $\LaTeX$, C++, Shell
- **Frameworks/Tools:** PyTorch, TensorFlow, Keras, Scikit, OpenCV, Numpy, Pandas, Git, Unity3D, vLLM, LangChain, FAISS, OpenAI API, Hugging Face Transformers & APIs